Calcurion Maths Worksheets

A-Level Mathematics

Topic: Pure 2 – Further Integration

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. $\int x^2 \ln x \, \mathrm{d}x$ (by parts). **[5]**

2. $\int xe^{2x} \, \mathrm{d}x$ (by parts). **[5]**

3. $\int x \sin x \, \mathrm{d}x$ (by parts). **[5]**

4. $\int \frac{2x}{1+x^2} \, \mathrm{d}x$ (substitution $u = 1 + x^2$). **[4]**

5. $\int \frac{3x^2}{x^3+1} \, \mathrm{d}x$. **[4]**

6. $\int \frac{5x+7}{(x+1)(x-2)} \, \mathrm{d}x$. **[5]**

7. $\int \frac{2x-1}{(x+1)(x-3)} \, \mathrm{d}x$. **[5]**

8. $\int \frac{x}{x^2+4} \, \mathrm{d}x$. **[3]**

9. $\int \frac{1}{x^2+9} \, \mathrm{d}x$. **[3]**

10. Volume of revolution of $y = \sqrt{x}$ about x-axis from $x = 1$ to $x = 4$. **[5]**

11. Volume of revolution of $y = x^2$ about x-axis from $x = 0$ to $x = 2$. **[4]**

12. $\int_0^{\pi/4} x \cos 2x \, \mathrm{d}x$. **[6]**

13. $\int_0^1 x e^x \, \mathrm{d}x$. **[5]**

14. $\int \frac{x+1}{x^2+2x+5} \, \mathrm{d}x$. **[5]**

15. $\int \sin^2 x \, \mathrm{d}x$. **[4]**

Section B – Hard (Q16–25)

16. $\int e^{2x} \sin 3x \, \mathrm{d}x$ (parts twice). **[7]**

17. $\int e^x \cos 2x \, \mathrm{d}x$ (parts twice). **[7]**

18. Volume when region under $y = x^2 + 1$ from $x = 0$ to 2 is rotated about y-axis. **[6]**

19. Volume when region under $y = \sqrt{4 - x^2}$ from $x = 0$ to 2 is rotated about x-axis. **[5]**

20. $\int_0^{\pi/2} x \sin x \, \mathrm{d}x$. **[6]**

21. $\int \frac{2x+3}{(x-1)^2(x+2)} \, \mathrm{d}x$ (partial fractions). **[7]**

22. $\int \ln x \, \mathrm{d}x$. **[4]**

23. $\int x^2 e^x \, \mathrm{d}x$ (parts twice). **[6]**

24. $\int \frac{1}{\sqrt{4-x^2}} \, \mathrm{d}x$. **[3]**

25. $\int_1^e \frac{\ln x}{x} \, \mathrm{d}x$. **[4]**

End of Worksheet • *Show all working clearly.*